

Gen-3 Consolidated Results: Status of the Full Neural Replacement of the LHCb RK Extrapolator

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Abstract

We consolidate three generations of LHCb track-extrapolation experiments against the canonical project goal: *replace* the C++ adaptive Runge–Kutta extrapolator with a neural network whose inference path contains no analytic Lorentz right-hand side and no magnetic-field lookup. Under this strict criterion, the best replacement candidate to date is `pinn_v2_small_v1` at $\langle |\Delta x| \rangle = 0.117$ mm on the signed- Δz test split, with 10 372 trainable parameters [source: `experiments/gen_3/README.md §M1 results`]. The previously-reported best, `nrk4_tiny_1step_v1` at 0.113 mm, is **demoted from the deployment slot** by ADR 0009 [11] on the grounds that it evaluates the analytic Lorentz right-hand side and the field map at inference and therefore *augments*, rather than *replaces*, the classical extrapolator. The `PROJECT_CONTEXT.md` minimum gate is < 0.10 mm position error (ideally < 0.05 mm) [1]; the current best replacement is $\sim 1.17\times$ above the minimum. The gap is closable through the six-phase roadmap (R1–R6) in `REPLACEMENT_PLAN.md` [2]. This document supersedes the framing – but not the numerical content – of `gen3_m1_results.tex` (2026-05-08) [7].

1 Goal and re-framing

Canonical goal. `PROJECT_CONTEXT.md` states verbatim:

“Replace the C++ adaptive Runge-Kutta track extrapolator with a faster neural network that maintains high accuracy.” [1]

Replacement criterion. A candidate is a true replacement iff, at inference time, the neural network is the *entire* mapping

$$(x, y, t_x, t_y, q/p, z_{\text{start}}, \Delta z) \longmapsto (x_f, y_f, t_{x,f}, t_{y,f}, q/p_f),$$

with **no field-map lookup** and **no analytic Lorentz step** on the forward path [2, 11]. Models that wrap a learned residual around a classical RK4 integrator of the analytic Lorentz ODE (the `NeuralRK4` family) are *hybrids*: they may be useful as a scaffold or fallback, but they do not satisfy the replacement criterion and must not occupy the deployment slot.

Re-framing relative to `gen3_m1_results.tex`. The previous results document (`gen3_m1_results.tex`, dated 2026-05-08 [7]) presented `nrk4_tiny_1step_v1` as the Stage-1 deployment candidate on the basis of its $\langle |\Delta x| \rangle = 113 \mu\text{m}$ and its single-model passage of all four Allen ABI gates at fp32. The numerical content of that document is correct and is retained here unchanged; the *deployment recommendation* is overturned by ADR 0009. This document is the post-ADR-0009 consolidated view. The detailed claim-by-claim re-mapping is in §8.

2 Cross-generation results

Table 1 lists the headline numbers for every named candidate across the three generations, ordered by generation and then by mean position error. The *Replacement?* column applies the criterion of §1: a ✓ indicates that the inference path contains neither field-map nor analytic-Lorentz step; a × marks hybrids. The marker ~ flags “fixed- Δz only” gen-1 numbers that are not directly comparable to the variable- Δz test splits of gen-2 and gen-3.

Table 1: Cross-generation replacement-candidate inventory. Position error is the mean $|\Delta x|$ on the per-generation test split. Gen-1 numbers are on the *fixed* $\Delta z = 8000$ mm slice (marker ~) and are not directly comparable to gen-2/gen-3 variable- Δz numbers. The highlighted row is the current best true replacement.

Gen	Model	Family	Repl.?	Params	$\langle \Delta x \rangle$	Source
1	mlp_tiny_v1	MLP	✓	2 660	$\sim 22.8 \mu\text{m}$	[6]
1	mlp_xlarge_v1	MLP	✓	430 980	$\sim 27.4 \mu\text{m}$	[6]
2	mlp_tiny	MLP	✓	4 868	0.387 mm	[5]
2	mlp_medium	MLP	✓	102 788	0.305 mm	[5]
2	mlp_large	MLP	✓	401 412	0.380 mm	[5]
2	pinn_v2_medium_lam0.1	PINN_v2	✓	68 612	0.235 mm	[5]
2	pinn_v2_medium_lam1.0	PINN_v2	✓	68 612	0.322 mm	[5]
2	neural_rk4_small_1step	NeuralRK4 (hybrid)	×	17 925	0.125 mm	[5]
3	mlp_small_v1_broken	MLP	✓ (broken)	18 076	18.41 mm	[4]
3	mlp_medium_v1_broken	MLP	✓ (broken)	51 076	5.57 mm	[4]
3	pinn_v2_small_v1	PINN_v2	✓ (best)	10 372	0.117 mm	[4]
3	_archive_hybrid/nrk4_tiny_1step_v1	NeuralRK4 (hybrid)	×	4 997	0.113 mm	[4]
3	_archive_hybrid/nrk4_small_1step_v1	NeuralRK4 (hybrid)	×	18 181	0.210 mm	[4]

Reading the table. Three observations.

1. Gen-1 MLPs reached tens of μm mean position error *on a fixed- Δz slice with no z_0 dependence* [6]. That regime is not representative of Allen’s call pattern and the gen-1 numbers must not be quoted out of context.
2. Gen-2 demonstrated for the first time that a true replacement (**pinn_v2_medium_lam0.1**, 0.235 mm) is within $\sim 2\times$ of the best hybrid (**neural_rk4_small_1step**, 0.125 mm) on the variable- Δz contract [5].
3. Gen-3 closes that gap entirely: **pinn_v2_small_v1** at 0.117 mm sits *below* the best gen-2 hybrid and is statistically indistinguishable from the best gen-3 hybrid (§3) [4].

3 Headline result: PINN_v2 vs hybrid in gen-3

The two best gen-3 candidates report [4]:

pinn_v2_small_v1	0.117 mm mean $ \Delta x $,	10 372 parameters (40.5 kB fp32)
nrk4_tiny_1step_v1	0.113 mm mean $ \Delta x $,	4 997 parameters (19.5 kB fp32)

The difference of $\approx 4 \mu\text{m}$ on a $\sim 117 \mu\text{m}$ baseline is inside the noise budget of the F2 heavy-tail artefact (§4.2) [4]: the same **nrk4_tiny_1step_v1** checkpoint measured on the train/val/test splits gives $|\Delta x|$ values of 125/122/113 μm respectively – a spread of 12 μm on the same model and the same metric. The 4 μm gap between PINN_v2 and NRK4 is smaller than that intra-model split-to-split variability and is therefore not significant.

Operational conclusion. The principal reason to retain the hybrid – a clear accuracy advantage at matched parameter count – *has evaporated under the gen-3 data contract*. A true replacement is competitive with the best hybrid today, with only $\sim 17\%$ of headline error to close to reach the PROJECT_CONTEXT.md < 0.10 mm minimum gate. The hybrid is no longer the deployment candidate [11, 3].

4 What is known to be broken and why

Three quantitative pathologies block the current best replacement from clearing the < 0.10 mm gate. Each has a specific fix scheduled in REPLACEMENT_PLAN.md §4 [2].

4.1 MLP collapse (Fix M1)

The gen-3 MLP recipe was inherited verbatim from gen-2 with a 6-dim input $[x, y, t_x, t_y, q/p, \Delta z]$. The gen-2 contract was forward-only ($\Delta z > 0$) and depth-fixed, so the input was sufficient. The gen-3 contract is signed $\Delta z \in [-10\,000, +10\,000]$ mm with z_0 sampled across the full LHCb z -range [4]. With no sign or z_0 input, the network cannot distinguish forward from backward extrapolation and cannot account for the $\sim 15\times$ variation of $|B_y|$ between the fringe and the field-map peak.

The empirical signature is a 1–2 order-of-magnitude regression relative to the corresponding gen-2 MLP [5, 4]:

$$\underbrace{0.305 \text{ mm}}_{\text{mlp_medium (gen-2)}} \longrightarrow \underbrace{5.57 \text{ mm}}_{\text{mlp_medium_v1_broken (gen-3)}} \sim 18 \times .$$

Fix M1 [2] prescribes a 7-dim input $[x, y, t_x, t_y, q/p, z_{\text{start}}, \Delta z]$ plus engineered features $\log_{10}(|\Delta z|/100)$ and $\text{sign}(\Delta z)$, training on the full 10 M corpus rather than the 200 k subset used in M1. The expected outcome is recovery to ~ 0.3 mm at minimum, ideally ~ 0.15 mm.

4.2 Loss heavy-tail (Fix L1)

Quoted directly from gen-3 README §F2 [4]: running the same `nrk4_tiny_1step_v1` checkpoint through the same `validate()` function on each split gives

Split	loss (squared, normalised)	$ \Delta x $ (mm)	pos_rms (mm)
val	1.426×10^{-5}	0.122	1.15
test	1.918×10^{-7}	0.113	0.66
train	2.162×10^{-4}	0.125	2.64

The loss varies by a factor of $1100\times$ between splits while mean $|\Delta x|$ varies by less than 10%. The cause is a heavy-tail in the per-track residual: $\ll 1\%$ of high- $|q/p|$, large- $|\Delta z|$, backward-extrapolated tracks each contribute $\sim 10^4\times$ the weight of a typical track under squared-MSE, and the count of such tracks fluctuates between splits.

Implication. The squared-MSE loss is **not** a valid quantitative selection metric on the signed- Δz contract [4]. Fix L1 [2] replaces it with log-cosh training loss and median- $|\Delta x|$ selection; all R1–R6 results will be reported with mean, median, 68%, 95%, 99% quantiles of $|\Delta x|$ and $|\Delta \text{slope}|$.

4.3 PINN_v2 parameter starvation (Fix P1)

`pinn_v2_small_v1` has 10 372 trainable parameters and was trained on a 200 k-track subset of the 10 M corpus [4]. The gen-2 PINN_v2 family was at 68 612 parameters and trained on the full 50 M; the closest analogue across the data-contract change is therefore severely under-scaled. Fix P1 [2] sweeps `hidden_dims` $\in \{[256, 256], [256, 256, 128], [512, 256, 256]\}$ on the 10 M corpus (largest variant ~ 200 k parameters) at the gen-2-best $\lambda_{\text{pde}} = 0.1$, with a late-stage λ_{pde} decay to 0.01 over the final 20% of training. The target is 0.08–0.10 mm at ~ 100 –200 k parameters, which would clear the PROJECT_CONTEXT.md minimum gate [1].

5 What is known to be sound

5.1 Pure RK4 baseline on the gen-3 7-input contract

Pure classical RK4 of the analytic Lorentz ODE on the gen-3 formulation – i.e. `nrk4_tiny_1step_v1` evaluated at epoch 1 *before any learning has occurred* – delivers $\langle |\Delta x| \rangle = 47 \mu\text{m}$ on the test set with **zero trainable parameters** [4]. This is the gen-3 reference baseline against which all candidates must be measured. It is *not* a deployment candidate: it is exactly the C++ classical-RK code we are replacing, instantiated in Python.

5.2 fp64-autograd Jacobian methodology (ADR 0007)

The original “structural A4 failure” that motivated multi-step hybrid integrators [10] was, on re-examination, an **fp32 + finite-difference numerical artefact**: with $\varepsilon \sim 10^{-3}$ on a fp32 model evaluated through a multi-metre propagation, the finite-difference signal in the small off-diagonal Jacobian entries sits at the fp32 round-off floor ($\sim 10^{-7}$ of the output magnitude), and random sign noise produces $\sim 100\%$ relative error. Re-running with the model cast to fp64 and the Jacobian computed by autograd gives a Frobenius relative error of 7×10^{-9} against the RK4 reference [10].

This unlocks the use of A4 as a real quantitative gate on the *replacement* candidates. The corresponding re-measurement across MLP and PINN_v2 – using the same fp64-autograd routine lifted into a generic `evaluate_a4(model, test_states)` utility – is Phase R2 of the roadmap [2].

5.3 V3 weight-blob export pipeline

The existing `For_Allen/scripts/export_bin.py` pipeline (V2 format on disk, V3 format specified in `For_Allen/docs/decisions/` and the M1.5 audit) is format-compatible across the MLP, PINN_v2, and NeuralRK4 families with only a small architecture-tag change in the header [4, 3]. The header magic becomes a generic NN_EXTRAP marker with an `architecture` field $\in \{\text{MLP}, \text{PINN_V2}, \text{NRK4}\}$; the per-layer record layout, the fp32/fp16 quantisation path, and the CRC32/git-hash footer all carry over unchanged. Phase R5 is therefore a small follow-up MR, not a re-implementation.

6 Gates against PROJECT_CONTEXT.md

Table 2 maps the PROJECT_CONTEXT.md success criteria, plus the seven Allen integration constraints A1–A7 inherited from the For_Allen audit, onto the current best measured value. The table is the unambiguous answer to “where do we stand against the project’s own success criteria today.”

Status summary. Three gates are unmeasured (A4, A6, and the GPU-cost row of the PROJECT_CONTEXT block). One is fail (the $< 0.05\text{mm}$ ideal). One is close (the $< 0.10\text{mm}$ minimum, at $1.17\times$ the threshold). Six are pass. There is no gate that is known to be unreachable.

7 What is next

The six-phase roadmap to deployment is REPLACEMENT_PLAN.md §6 [2]. Each phase produces a specific artefact; nothing proceeds without a written verdict from the preceding phase.

Phase R1 – loss metric reform. Change `models/train.py` to log-cosh loss and `models/eval.py` to report median, 68%, 95%, 99% quantiles of $|\Delta x|$ and $|\Delta \text{slope}|$; change all selection criteria from `val_loss` to `median_dx_mm`. Re-evaluate (no retraining) the M1 checkpoints on the new metric. *Artefact*: a frozen table of mean/median/95%/99% numbers for `mlp*_v1`, `pinn_v2_small_v1`, and the archived hybrid checkpoints.

Table 2: Gates against PROJECT_CONTEXT.md and the Allen integration constraints. “Unmeasured” = the corresponding fp64-autograd measurement has not been performed on the replacement candidates since the methodology was unlocked (see §5.2); it is scheduled for Phase R2.

Gate (source)	Threshold	Current value	Status
Mean $ \Delta x $ minimum [1]	< 0.117 mm	(<code>pinn_v2_small_v1</code>) [4] 0.10 mm	close
Mean $ \Delta x $ ideal [1]	< 0.117 mm	(same) 0.05 mm	fail
Per-track GPU cost [1]	fastest	not yet measured on replacement candidates than ~ 2.5 μ s C++ RK4	unmeasured
Variable Δz support [1]	full	supported in gen-3 data contract signed $\Delta z \in [-10^4, 10^4]$ mm	pass
Allen integrable (<code>ITrackExtrapolator</code>) [1]	drop-MLP family already wired via <code>TrackMLPExtrapolator.cpp</code> in API		pass (
A1 constant memory [2]	no	both MLP and PINN_v2 satisfy by construction per-event alloc	pass
A2 fp32/fp16 round-trip [2]	bit-exact	existing V3 pipeline reload	pass
A3 weight blob [2]	≤ 40.5 kB	(<code>pinn_v2_small_v1</code> , fp32) 64 kB	pass
A4 Jacobian agreement [2, 10]	Prob. not > 0.05, off-diag rel. < 0.20	not measured on replacement candidates with fp64-autograd	unmeasured
A5 fp32 inference bit-bound [2]	$\leq 10^{-4}$	MLP path proven; PINN_v2 shares arithmetic rel. vs Python	pass
A6 GPU per-track cost [2]	$\leq$	not yet measured classical RK4 cost	unmeasured
A7 API drop-in [2]	no	<code>TrackMLPExtrapolator</code> signature reusable Moore/Allen API change	pass

Phase R2 – A4 Jacobian re-measurement. Lift the fp64-autograd Jacobian routine from `phase1a_arch_ablation.py` [10] into a generic `evaluate_a4(model, test_states)` utility; run

it on `mlp_medium_v1` (broken), `pinn_v2_small_v1`, and the cached fp64 RK45 reference Jacobian. *Artefact*: pass/fail verdict on A4 for the two replacement candidates. If either passes, true-replacement feasibility is empirically established; if neither passes, escalate to Phase R-X (Jacobian co-supervision or straight-line residual output) [2].

Phase R3 – MLP modernisation. Change `models/architectures.py::MLP` input to the 7-dim contract plus the two engineered features of §4.1; train `mlp_{small,medium,large}_v2` on the 10 M corpus, 60 epochs, log-cosh loss. *Artefact*: at least one MLP at < 0.5 mm median $|\Delta x|$ with A3-compliant parameter budget, or a documented retirement of the MLP family.

Phase R4 – PINN_v2 scaling. Three-way width sweep $\times$ 10 M corpus, 60 epochs, log-cosh loss, cosine-decayed λ_{pde} as in §4.3. *Artefact*: at least one PINN_v2 at < 0.10 mm median $|\Delta x|$ that also passes A4 – the deployment candidate.

Phase R5 – bin export + smoke-test bridge. Re-anchor `For_Allen/PLAN.md` onto the R4 winner; re-purpose `NRKExtrapolator.cuh` in Allen MR !2497 so that the inner loop calls a generic `MLForwardPass(weights, state)` rather than the RKN4 ladder; re-run the bit-bound smoke test against the new candidate. *Artefact*: V3 weight blob, `NeuralTrackExtrapolator.cuh`, smoke-test verdict $\|y_{\text{py}} - y_{\text{cpp}}\|_{\infty} < 10^{-6}$ mm in x, y .

Phase R6 – Allen MR + Moore integration. Standard Phases 5–8 of the existing `For_Allen/PLAN.md`: throughput test in `hlt1_pp_default` (A6), Moore physics test acceptance (A4, A5), MR review. *Artefact*: merged Allen MR and a Moore physics-test PR.

8 Reconciliation with prior documents

This section is the explicit re-mapping between the abstract of `gen3_m1_results.tex` (2026-05-08) [7] and the post-ADR-0009 view. **The numerical content of the prior document is unchanged.** The *deployment recommendation* is overturned. Table 3 is claim-by-claim.

One-line takeaway. The `gen3_m1_results.tex` numbers are correct; its *deployment recommendation* was based on a criterion (best ABI-compliant model irrespective of architecture class) that has since been tightened to the replacement criterion of §1 [11]. Under the tightened criterion the deployment candidate is the R4 PINN_v2 winner, not `nrk4_tiny_1step_v1`.

9 Open questions and risks

Reproduced from `REPLACEMENT_PLAN.md` §8 [2]:

1. **Should we retain a small NeuralRK4 fallback in Allen?** Pro: zero-risk safety net behind a property switch. Con: maintenance burden, confuses the deployment story. *Recommendation*: keep the C++ surface (`NRKExtrapolator.cuh`), remove the trained Python NRK4 from the production candidate list.
2. **Is the 10 M corpus sufficient at the widest PINN_v2 variant?** Unknown: gen-3 has never trained [512, 256, 256]. If validation curves are still descending at epoch 60 of Phase R4, generate 20 M.
3. **Universal vs station-specialised models?** A single universal model on signed $\Delta z \in [-10\,000, 10\,000]$ mm is the cleanest test; the 0.117 mm gen-3 result on universal Δz is the most stringent benchmark. Default to universal until evidence forces a $z_{\text{end}} \in \{\text{VELO}, \text{UT}, \text{SciFi}\}$ split.

Table 3: Claim-by-claim reconciliation of `gen3_m1_results.tex` (2026-05-08) abstract with the post-ADR-0009 view (this document).

Claim in <code>gen3_m1_results.tex</code> abstract	Numerical	Updated status
“ <code>nrk4_tiny_1step_v1</code> achieves $\langle \Delta x \rangle \approx 113 \mu\text{m}$ at ~ 4997 trainable parameters.”	correct	Number stands. Demoted from M1 winner because it is a hybrid [11].
“Two orders of magnitude better than either pure-MLP control.”	correct	Number stands. The MLP controls were <i>broken</i> (§4.1); the 2-OOM gap is against a pathological MLP, not against the MLP family itself. Phase R3 will retest.
“Of the five candidates, only <code>nrk4_tiny_1step_v1</code> satisfies all four hard ABI gates (A2, A3, A5, A7) at fp32; two more become compliant at fp16.”	correct	Number stands. Irrelevant to the deployment slot: ABI compliance of a hybrid does not make the hybrid a replacement.
“None of the five reaches the Stage-1 <i>physics</i> tolerance of $ \Delta x < 12 \mu\text{m}$ at the VELO exit.”	correct	Number stands. The Stage-1 $12 \mu\text{m}$ gate corresponds to the <code>PROJECT_CONTEXT.md</code> $< 0.05 \text{ mm}$ ideal target [1] and remains the stretch goal for Phase R4 / a follow-up campaign.
“M1 is architecturally green-lit but performance red-lit.”	—	Re-cast: M1 <i>architecturally surfaced</i> that the best true replacement (<code>PINN_v2</code> at 0.117 mm) is already competitive with the best hybrid (<code>NRK4</code> at 0.113 mm), at one fewer order of free parameters in the design space. The performance red-light is real (§6) and is the target of Phases R3–R4.
“Eight-action M1.5 backlog (M–T) targeting heavy-tail, <code>NRK4</code> backward asymmetry, <code>PINN</code> q/p bias, and the 10 M corpus move.”	—	Items M , P (loss reform, 10 M training) are subsumed by Phases R1, R3, R4 [2]. Items N (<code>NRK4</code> corrector outside <code>_rhs</code>) and Q (<code>NRK4</code> weight-blob export) are <i>archived</i> with the hybrid family [3]; the underlying <code>NRK4</code> work continues only as a research baseline. Items O (<code>PINN</code> refactor) and R , S , T (Allen kernel, OOD test, MLflow tracking) carry into the R-roadmap unchanged.

Companion theory document. A parallel document, `gen3_replacement_theory_2026-05-19.tex`, develops the theoretical case for each fix (L1, M1, P1) and for the Phase R-X contingency routes. The present document is the empirical inventory; the theory companion is the analytical justification.

References

- [1] `PROJECT_CONTEXT.md`, canonical project goal, April 2026. [TrackExtrapolation/PROJECT_CONTEXT.md](#).
- [2] `REPLACEMENT_PLAN.md` §3, §4, §6, §8, 2026-05-19. [experiments/gen_3/REPLACEMENT_PLAN.md](#).
- [3] `CLEANUP_LIST.md` (hybrid demotion), 2026-05-19. [experiments/gen_3/CLEANUP_LIST.md](#).

- [4] *Gen-3 experiments README.md* § “Headline numbers”, § “Diagnoses (F1, F2, F3)”, § “Likely thesis-section structure”, 2026-05-06. [experiments/gen_3/README.md](#).
- [5] *Gen-2 experiments README.md* § “v2 sweep” results table. [experiments/gen_2/README.md](#).
- [6] *Gen-1 evaluation report*, top-21 model ranking, 2026-01-15. [experiments/gen_1/archive/analysis_v2/evaluation_](#)
- [7] *Gen-3 Milestone-1 Track-Extrapolation Models* ([gen3_m1_results.tex](#)), 2026-05-08. [docs/reports/gen3_m1_results.tex](#).
- [8] *Gen-2 on New Data: v2 Architectural Fixes and Final Results* ([gen2_new_data_results.tex](#)). [docs/reports/gen2_new_data_results.tex](#).
- [9] *First-Generation Neural Track Extrapolation for LHCb Allen HLT1* ([gen1_results.tex](#)), April 2026. [docs/reports/gen1_results.tex](#).
- [10] ADR 0007 – *Phase 1a winner: $n_{rk_steps} = 2$, corrector OFF*; fp64-autograd Jacobian methodology, 2026-05-12. [For_Allen/docs/decisions/0007-phase1a-winner.md](#).
- [11] ADR 0009 – *Replacement goal restated; hybrid demoted*, 2026-05-19. [For_Allen/docs/decisions/0009-replacement-goal-restated.md](#).